

Combating Digital Misinformation: An AI-Powered Web Extension + Application

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I. Introduction

The rapid growth of social media and online platforms has made information easier to access and share than ever before. People can receive news, opinions, and other forms of information within seconds through websites, social media posts, videos, and messaging platforms. However, this convenience has also increased concerns about misinformation, making it difficult for users to determine whether the information they encounter is accurate or misleading. According to the Reuters Institute Digital News Report 2024, 59% of respondents across the 47 markets surveyed expressed concern about distinguishing what is real from what is fake when consuming news online (Newman et al., 2024).

Misinformation can have serious consequences because false or misleading information can cause confusion, influence decision-making, and reduce trust in reliable institutions. The World Health Organization (WHO) describes an “infodemic” as an overabundance of information, including false or misleading information, which can contribute to confusion, risk-taking behaviors, and mistrust, particularly during public health emergencies (The World Health Organization, 2022).

With the increasing availability of artificial intelligence (AI), technology can be used not only to create misleading content but also to assist users in evaluating information. Therefore, this project proposes an AI-powered browser extension and application that can help users check online posts and articles before believing or sharing them.

II. Problem Description

Misinformation has become a significant problem in today's digital environment because information can be produced and distributed rapidly without being properly verified. Social media platforms can expose users to information from unknown sources, influencers, personal accounts, and other users. The *Reuters Institute Digital News Report 2024* found that politics was the topic that generated the most concern about

false or misleading information, followed by health information and news about the wars in Ukraine and Gaza (Newman et al., 2024).

One major difficulty is that users may not have enough time, knowledge, or resources to independently verify every post or article they encounter. A person may trust an article because it appears professional, has many shares, or comes from someone they know. Furthermore, old news can be reposted as if it were current information, while edited images, misleading headlines, and AI-generated content can make false or misleading information appear convincing.

The problem is therefore not only the existence of misinformation but also the difficulty people face in quickly evaluating the credibility and context of online information. WHO recommends continuous monitoring of harmful and false online content and improving digital literacy as part of broader efforts to manage infodemics (The World Health Organization, 2022).

Media and information literacy is also important because users need the ability to critically evaluate online information. UNESCO emphasizes that media and information literacy helps individuals develop the critical-thinking skills needed to analyze and engage with online content and become more resilient to misinformation and other harmful content (UNESCO, 2023).

III. Proposed Solution

The proposed solution is an AI-powered misinformation detection ecosystem featuring both a browser extension and a companion application designed to help users evaluate online posts, articles, and digital content seamlessly. For desktop browsing, the extension allows users to analyze selected text or articles directly as they surf the web. For mobile and outside the browser, the application offers two convenient methods: users can either copy and paste a suspicious link directly into the app to trigger a verification check, or use the standard "Share" button in their mobile browser or social media apps to send the content straight to the app, which then pops up with the results. To assess accuracy, the system examines multiple factors, including publication dates, author and organization credibility, the wording of claims, supporting evidence, and cross-references with established fact-checking databases, government agencies, academic institutions, and reliable news sources. Rather than giving a simple

true or false label, the system categorizes content as "Likely Reliable," "Needs Verification," or "Likely Misleading/False," and provides a concise explanation alongside direct links to the sources used. This enables users to easily understand why a piece of content received its rating, such as warning them when an old article is being reshared as breaking news, while also empowering them to independently review the evidence.

IV. Conclusion

In an era where information spreads at unprecedented speeds, the ability to distinguish fact from fabrication is no longer just a digital convenience. It is a critical necessity. As highlighted by the ongoing challenges of the "infodemic," misinformation poses a direct threat to public health, individual decision-making, and societal trust. By providing a dual-platform solution that integrates an AI-powered browser extension with a mobile application, this project bridges the gap between the abundance of online data and the need for immediate, reliable verification. By empowering users to authenticate content through simple link-sharing and instant analysis, this tool fosters greater media literacy and critical thinking. As an individual in this technology-rich world, I recognize that accurate information is the foundation of well-informed health decisions; therefore, this technology serves as a vital safeguard, ensuring that users can navigate the digital landscape with confidence, resilience, and a clearer understanding of the information they consume and share.

V. References

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364

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